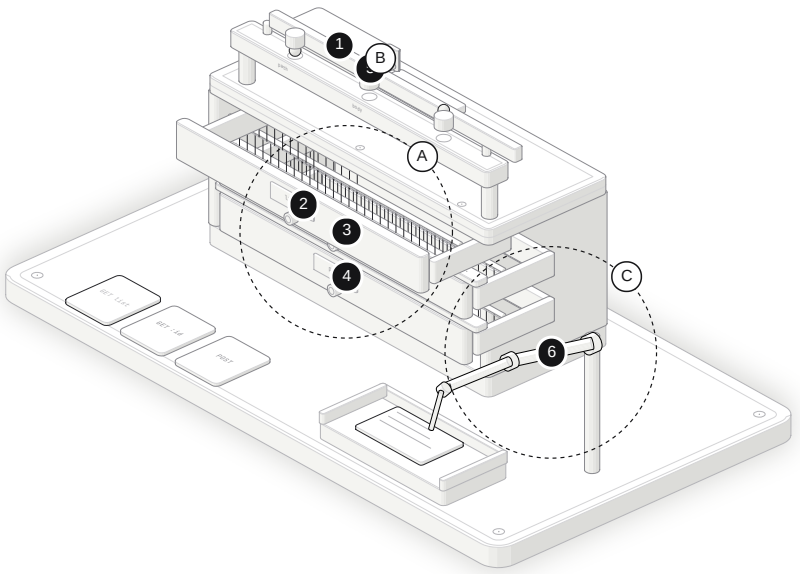
Mapper



How does a plain C# method become an HTTP operation?

Every front door — ServiceResource, Controller or an Inline lambda — ends in the same machinery: TypeAnalyzer reads the attributes into a MethodCollection and the OperationRouter matches the request card to one drawer. SignatureAnalyzer then binds each parameter by fixed precedence, so a bead drops only into the slot the signature declares.

3 front doors
SERVICERESOURCE, CONTROLLER, INLINE

6 slots
FIXED BINDING PRECEDENCE: PATH TO CONTENT

15 files
A MINI-MEDIATR-PLUS-MVC, NO CODE-GEN STEP

2 paths
ROSLYN-COMPILED, OR REFLECTION FALLBACK

403 intercepted
[REQUIREROLE] BLOCKS BEFORE THE METHOD

PARTS

01 **The request card**
Method and path, dropped into the throat to be matched.

02 **Drawer — GET list**
One operation per method, read from the attributes.

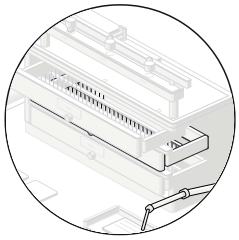
03 **Drawer — GET :id**
Literal, :variable and regex segments all route here.

04 **Drawer — POST**
A body-carrying operation; the tray takes its answer.

05 **The beads**
Bound arguments, seated by precedence, never by guesswork.

06 **The pen arm**
The Roslyn-compiled invocation, writing the response.

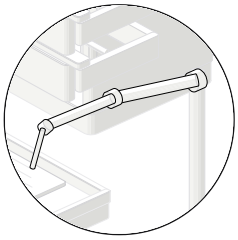
CLOSE-UPS



(A) **A drawer**
One operation: route segments parsed at build time, wildcards closing into sub-handlers when a method returns a handler.

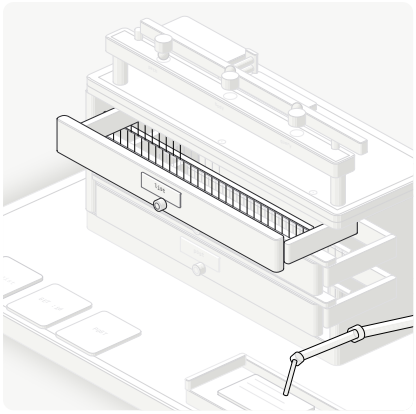


(B) **The beads**
Arguments bound by fixed precedence. Injectors cover the request, the handler, the auth user and any registered service.



(C) **The pen**
Compiled once per operation, invoked per request — with the interceptor as the guard who can stop the hand mid-air.

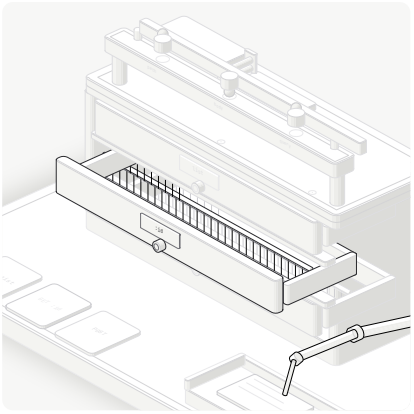
How it works



01

Three doors, one engine

ServiceResource, Controller or Inline.Create(): every front door ends in MethodCollection, OperationRouter, MethodHandler.



02

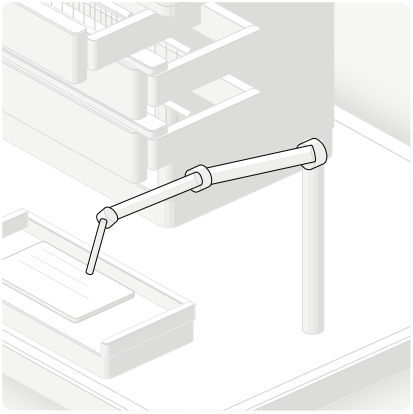
Match the card

The OperationRouter reads the attributes once and matches the request to one drawer — segments as literals, :variables or regexes.

03

Bind by precedence

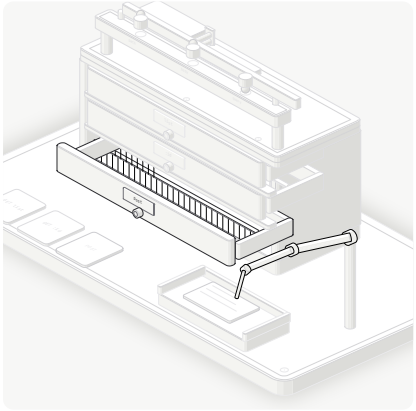
Path, injected, streamed, body, query, content: SignatureAnalyzer seats each argument in the slot its signature declares.



04

The pen writes

ExecutionMode.Auto compiles the call with Roslyn into a real delegate — and falls back to plain reflection where it cannot.



05

Interception first

An interceptor such as [RequireRole("admin")] can short-circuit the call before the method runs — 401 or 403, documented.

WHY COMPILED?

MethodInfo.Invoke per request is the classic reflection tax. ExecutionMode.Auto instead emits a real Invoke(...) with Roslyn and compiles it once per operation, so the hot path is a plain call — falling back to reflection only where Roslyn cannot run, or for internal methods, deliberately.

SPECIFICATION

Analyzer	TypeAnalyzer, public and internal	Fast path	ExecutionMode.Auto, Roslyn delegate
Segments	literal, :variable, regex, closing	Formats	JSON default; YAML, XML, Protobuf addable
Binding	path, injected, streamed, body, query, content	Interceptors	[InterceptWith<T>], OpenAPI-documented
Injectors	request, handler, user, services, body		